Table 20: Performance and unparseable rates for few-shot direct answer, few-shot CoT, Plan + Direct Solver, Plan + CoT Solver, and Plan + Tool Solver Solver. "Acc." stands for accuracy and "% Unp." stands for the rate of unparseable model responses that either fail to pass our answer extraction parser (for all methods except Plan + Tool Solver prompting) or fail to be executed by symbolic solvers. For FOLIO and ContextHub, we compute the accuracy by making a random guess for the unparseable responses; for GSM8K and GSM8K-Hard, we consider the unparseable responses as incorrect.

Dataset	Method	Mistral 7b		Llama 3.1 8b		Llama 3.1 70b		GPT-4o Mini	
		Acc.	% Unp.	Acc.	% Unp.	Acc.	% Unp.	Acc.	% Unp.
GSM8K	Direct Answer	12.5	0.1	20.1	0.5	39.1	0.0	32.8	0.0
GSM8K	CoT	56.2	1.4	86.4	1.0	96.1	0.1	94.2	0.1
GSM8K	Plan + CoT Solver	45.0	1.0	78.7	0.4	94.7	0.0	92.0	0.1
GSM8K	Plan + Direct Solver	10.6	0.1	19.6	0.1	42.2	0.0	39.3	0.0
GSM8K	Plan + Tool Solver	59.8	8.6	80.3	1.3	94.4	0.4	90.5	1.5
GSM8K-Hard	Direct Answer	2.9	0.7	4.4	0.6	12.8	0.7	12.3	7.6
GSM8K-Hard	CoT	20.3	5.0	32.4	9.6	47.8	4.4	52.2	0.5
GSM8K-Hard	Plan + CoT Solver	18.7	2.6	32.4	1.3	49.7	0.6	51.5	0.3
GSM8K-Hard	Plan + Direct Solver	3.0	0.5	5.5	0.8	15.8	0.1	17.4	0.3
GSM8K-Hard	Plan + Tool Solver	44.2	8.9	57.9	1.2	68.0	0.5	70.4	1.4
ContextHub Deductive L1	Direct Answer	59.2	2.8	23.0	0.0	50.0	0.0	44.3	0.0
ContextHub Deductive L1	CoT	46.2	0.2	73.0	0.2	67.5	0.0	59.2	0.0
ContextHub Deductive L1	Plan + CoT Solver	49.5	0.0	64.8	0.0	65.5	0.0	63.2	0.0
ContextHub Deductive L1	Plan + Direct Solver	45.8	3.0	55.8	0.0	53.5	0.0	56.2	0.0
ContextHub Deductive L1	Plan + Tool Solver	68.8	27.8	84.2	11.8	91.7	9.8	90.7	7.8
ContextHub Abductive L1	Direct Answer	21.7	2.8	36.1	0.0	58.9	0.0	59.2	0.0
ContextHub Abductive L1	CoT	23.9	0.0	40.0	0.0	62.2	0.0	76.9	0.0
ContextHub Abductive L1	Plan + CoT Solver	38.3	0.0	42.5	0.0	65.6	0.0	74.2	0.0
ContextHub Abductive L1	Plan + Direct Solver	46.9	3.9	33.3	0.3	63.1	0.0	61.7	0.0
ContextHub Abductive L1	Plan + Tool Solver	59.2	35.8	70.8	9.7	73.9	4.2	74.7	10.3
FOLIO	Direct Answer	56.2	12.3	59.6	0.0	69.5	0.0	64.0	0.0
FOLIO	CoT	53.7	1.5	56.7	2.5	72.4	2.0	70.4	0.0
FOLIO	Plan + CoT Solver	53.7	0.0	55.7	0.0	73.9	0.5	70.4	0.0
FOLIO	Plan + Direct Solver	52.7	0.0	54.2	0.0	72.9	0.0	63.5	0.0
FOLIO	Plan + Tool Solver	48.8	46.8	54.2	28.6	70.0	16.7	62.6	25.1

formal specifications that fail to follow the format of the formal language (Python or z3) and that lead to execution errors. Such an issue is particularly severe for the smaller models. However, we note that despite the high unparseable rate, the overall accuracy of these models with tool augmentation is still on par with or outperforms other methods.

I Discussion of Limitations

I.1 Long Horizon Planning

One set of tasks where symbolic reasoning helps substantially that our experiments haven't covered as thoroughly (with the exception of BiGGen-Bench) is long-horizon planning (Valmeekam et al., 2023; Xie et al., 2024; Gundawar et al., 2024; Valmeekam et al., 2024). There are two reasons we don't treat it here. First, we are primarily interested in tasks that are conveyed in language, and we see less complex planning in language-only tasks. Second, there has already been a large debate on the effectiveness of CoT, both pro (Huang et al., 2022; Hu et al., 2023) and against (Valmeekam et al., 2023; Kambhampati, 2024; Kambhampati et al., 2024b; Stechly et al., 2024a; Guan et al., 2024; Verma et al., 2024; Gundawar et al., 2024; Stechly et al., 2024b) using CoT and its derivatives like tree-of-thought (Yao et al., 2023; Kang et al., 2024), that has resulted in complex systems to help solve planning problems better. While story generation and interpretation involve elements of planning with natural language (Peng et al., 2022; Karpinska et al., 2024), such tasks are not conventionally formalized and benchmarked as planning and reasoning.

I.2 Dataset contamination

One limitation of our study is the presence of possible data contamination: it is unknown which benchmarks may have been explicitly pre-trained on by language models. If a model had memorized answers to benchmark questions, we would expect direct answering to close some of the gap with CoT, as the model can just reproduce a known answer rather than deriving it from scratch. We argue there are four reasons that our general conclusions are still trustworthy. First, we use a range of language model scales, including small models that have less capacity to memorize. Second, datasets with poor direct answering performance like GSM8K-Hard are unlikely to have been substantially memorized. Third, the inclusion of recent datasets such as MuSR (Sprague et al., 2024) and BiGGen Bench (Kim et al., 2024) helps to defray this risk. Fourth, our survey of the literature includes papers that were submitted to conferences in 2023, representing a range of older LLMs trained at various times.

J EXAMPLE PROMPTS

We will release all prompts and model responses on our Huggingface repo. We list a few prompt response pairs here in this section.

AGIEval LSAT AR zero-shot CoT prompt for Llama 3.1 70B

```
<|start_header_id|>user<|end_header_id|>

Explain your reasoning step-by-step for each question before answering.
Give your final answer in the format \"The answer is therefore <A, B, C,
D, E>\". Failure to comply with the answer formatting will result in no
credit.
Of the eight students\u2014George, Helen, Irving, Kyle, Lenore, Nina,
Olivia, and Robert\u2014in a seminar, exactly six will give individual
oral reports during three consecutive days\u2014Monday, Tuesday, and
Wednesday. Exactly two reports will be given each day\u2014one in the
morning and one in the afternoon\u2014according to the following
conditions: Tuesday is the only day on which George can give a report.
Neither Olivia nor Robert can give an afternoon report. If Nina gives a
report, then on the next day Helen and Irving must both give reports,
unless Nina's report is given on Wednesday.Q: Which one of the following
could be the schedule of the students' reports? Answer Choices: (A)Mon.
morning: Helen; Mon. afternoon: Robert Tues. morning: Olivia; Tues.
afternoon: Irving Wed. morning: Lenore; Wed. afternoon: Kyle (B)Mon.
morning: Irving; Mon. afternoon: Olivia Tues. morning: Helen; Tues.
afternoon: Kyle Wed. morning: Nina; Wed. afternoon: Lenore (C)Mon.
morning: Lenore; Mon. afternoon: Helen Tues. morning: George; Tues.
afternoon: Kyle Wed. morning: Robert; Wed. afternoon: Irving (D)Mon.
morning: Nina; Mon. afternoon: Helen Tues. morning: Robert; Tues.
afternoon: Irving Wed. morning: Olivia; Wed. afternoon: Lenore (E)Mon.
morning: Olivia; Mon. afternoon: Nina Tues. morning: Irving; Tues.
afternoon: Helen Wed. morning: Kyle; Wed. afternoon: George
Let's think step by
step.<|eot_id|><|start_header_id|>assistant<|end_header_id|>
```